

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- Gaussian $N(\mu, \sigma^2)$: 표준화 $Z=(X-\mu)/\sigma \sim N(0,1)$, Φ 표, 68-95-99.7
- 결합 $f(x,y)$, 주변 $f_X = \int f \, dy$, 곱셈규칙 $f(x,y)=f(x|y)f_Y(y)$, 독립 $\Leftrightarrow$ 분리

문항 목차 (총 12문항)

- [1] L06 · IPSRP 5.4.16 — 연속RV2
- [2] L06 · IPSRP 5.4.20 — 연속RV2
- [3] L06 · IPSRP 5.4.21 — 연속RV2
- [4] L06 · PSP 4.5.1 — 연속RV2
- [5] L06 · PSP 4.5.2 — 연속RV2
- [6] L06 · PSP 4.5.5 — 연속RV2
- [7] L06 · PSP 4.5.10 — 연속RV2
- [8] L06 · PSP 4.5.13 — 연속RV2
- [9] L06 · PSP 5.1.1 — 연속RV2
- [10] L06 · PSP 5.1.2 — 연속RV2
- [11] 복습 L02 · PSP 1.4.1 — 조건부·Bayes
- [12] 복습 L02 · PSP 2.2.1 — 조건부·Bayes

Problem 16

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} \frac{1}{2}e^{-x} + \frac{cy}{(1+x)^2} & 0 \leq x, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c .
- Find $P(0 \leq X \leq 1, 0 \leq Y \leq \frac{1}{2})$.
- Find $P(0 \leq X \leq 1)$.

Problem 20

Let $X \sim N(0, 1)$.

- a. Find the conditional PDF and CDF of X given $X > 0$.
- b. Find $E[X|X > 0]$.
- c. Find $\text{Var}(X|X > 0)$.

Problem 21

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} x^2 + \frac{1}{3}y & -1 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

For $0 \leq y \leq 1$, find the following:

- The conditional PDF of X given $Y = y$.
- $P(X > 0 | Y = y)$. Does this value depend on y ?
- Are X and Y independent?

4.5.1 ● Y is a continuous uniform $(1, 5)$ random variable.

(a) What is $P[Y > E[Y]]$?

(b) What is $P[Y \leq \text{Var}[Y]]$?

4.5.2● The current Y across a $1\text{ k}\Omega$ resistor is a continuous uniform $(-10, 10)$ random variable. Find $P[|Y| < 3]$.

4.5.5● The time delay Y (in milliseconds) that your computer needs to connect to an access point is an exponential random variable.

- (a) Find $P[Y > E[Y]]$.
- (b) Find $P[Y > 2 E[Y]]$.

4.5.10■ X is a continuous uniform $(-5, 5)$ random variable.

- (a) What is the PDF $f_X(x)$?
- (b) What is the CDF $F_X(x)$?
- (c) What is $E[X]$?
- (d) What is $E[X^5]$?
- (e) What is $E[e^X]$?

4.5.13 ■ The probability density function of random variable X is

$$f_X(x) = \begin{cases} (1/2)e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $P[1 \leq X \leq 2]$?
- (b) What is $F_X(x)$, the cumulative distribution function of X ?
- (c) What is $E[X]$, the expected value of X ?
- (d) What is $\text{Var}[X]$, the variance of X ?

5.1.1 ● Random variables X and Y have the joint CDF

$$F_{X,Y}(x,y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & x \geq 0; \\ & y \geq 0, \\ 0 & \text{ow.} \end{cases}$$

- (a) What is $P[X \leq 2, Y \leq 3]$?
- (b) What is the marginal CDF, $F_X(x)$?
- (c) What is the marginal CDF, $F_Y(y)$?

5.1.2● Express the following extreme values of $F_{X,Y}(x,y)$ in terms of the marginal cumulative distribution functions $F_X(x)$ and $F_Y(y)$.

- (a) $F_{X,Y}(x, -\infty)$
- (b) $F_{X,Y}(x, \infty)$
- (c) $F_{X,Y}(-\infty, \infty)$
- (d) $F_{X,Y}(-\infty, y)$
- (e) $F_{X,Y}(\infty, y)$

1.4.1 ● Mobile telephones perform *handoffs* as they move from cell to cell. During a call, a telephone either performs zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). In addition, each call is either long (L), if it lasts more than three minutes, or brief (B). The following table describes the probabilities of the possible types of calls.

	H_0	H_1	H_2
L	0.1	0.1	0.2
B	0.4	0.1	0.1

- (a) What is the probability that a brief call will have no handoffs?
- (b) What is the probability that a call with one handoff will be long?
- (c) What is the probability that a long call will have one or more handoffs?

2.2.1● On each turn of the knob, a gumball machine is equally likely to dispense a red, yellow, green or blue gumball, independent from turn to turn. After eight turns, what is the probability $P[R_2Y_2G_2B_2]$ that you have received 2 red, 2 yellow, 2 green and 2 blue gumballs?